Project 3 Harmonic Analysis

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1 Introduction

Definition 1 (Weights)**.** *A* weight *is a function $w \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$ with $w \geq 0$. For a measurable set $E \subseteq \mathbb{R}^d$ we write $w(E) := \int_E w(x)\,dx$.*

Definition 2 (Weighted L^p spaces)**.** *Let w be a weight and let $1 < p < \infty$. We define*

$$L^p_w(\mathbb{R}^d) := \left\{ f : \mathbb{R}^d \to \mathbb{R} \text{ measurable} : \int_{\mathbb{R}^d} |f(x)|^p \, w(x)\,dx < \infty \right\},$$

with norm $\|f\|_{L^p_w} := \left(\int_{\mathbb{R}^d} |f(x)|^p \, w(x)\,dx\right)^{1/p}$.

The central object of this report is the Hardy–Littlewood maximal operator M, which assigns to each point the largest average of $|f|$ over cubes containing that point. In the proof of our main result we will also need the analogous operator in which these averages are taken with respect to the measure $w(x)\,dx$ instead of Lebesgue measure.

Definition 3 (Hardy–Littlewood maximal operator)**.** *Let w be a weight with $w(Q) > 0$ for every cube Q. For a measurable function f and $x \in \mathbb{R}^d$,*

$$M_w f(x) := \sup_{Q \ni x} \frac{1}{w(Q)} \int_Q |f(y)| \, w(y)\,dy, \tag{1}$$

where the supremum is taken over all cubes $Q \subset \mathbb{R}^d$ with sides parallel to the axes containing x. If the supremum is taken only over cubes Q centred at x, we write M^c_w. If $w \equiv 1$ we drop the subscript and write M and M^c.

It is well known that M is of weak type $(1,1)$ and strong type (p,p) on $L^p(\mathbb{R}^d)$ for $1 < p \leq \infty$ [3]. In the light of our $L^p_w(\mathbb{R}^d)$ definition a natural question arises, for fixed $1 < p < \infty$, for which weights w is the sublinear operator M bounded on $L^p_w(\mathbb{R}^d)$? In other words, for which w is there a constant C such that for all $f \in L^p_w(\mathbb{R}^d)$

$$\|Mf\|_{L^p_w} = \left(\int_{\mathbb{R}^d} (Mf)^p \, w\,dx \right)^{1/p} \leq C \left(\int_{\mathbb{R}^d} |f|^p \, w\,dx \right)^{1/p} = C\,\|f\|_{L^p_w}. \tag{2}$$

Remark 1 (The zero weight)**.** *If $w = 0$ almost everywhere, then $\|f\|_{L^p_w} = 0$ for every f, so $L^p_w(\mathbb{R}^d)$ is trivial and M is trivially bounded.*

Remark 2 (The endpoint $p = \infty$)**.** *For $p = \infty$, with the essential supremum taken with respect to $w(x)\,dx$, the answer is trivial: M is bounded if and only if $w > 0$ almost everywhere. If $w > 0$ a.e., then $w(x)\,dx$ and dx have the same null sets, so $\|Mf\|_{L^\infty} \leq \|f\|_{L^\infty}$ applies. If $w = 0$ on a set Z with $|Z| > 0$, then $f = \chi_Z$ has weighted supremum 0, while $Mf > 0$ everywhere. Hence we restrict to $1 < p < \infty$.*

Suppose that (2) holds for a weight w. A first observation is that w cannot vanish on a set of positive measure. The indicator function of a bounded part of such a set has norm zero in L^p_w, while its maximal function is positive everywhere, which contradicts (2). More generally, w cannot be much smaller on part of a cube than on the rest of it. A function concentrated where w is small has a small norm in L^p_w, while its maximal function is large on the whole cube, where w is large. Making this precise leads to the following necessary condition.

Proposition 1. *Let $1 < p < \infty$ and let w be a weight that is not almost everywhere zero. If (2) holds with constant C, then for every cube Q we have $\int_Q w^{-\frac{1}{p-1}}\,dx < \infty$ and*

$$\left(\frac{1}{|Q|}\int_Q w\,dx\right)\left(\frac{1}{|Q|}\int_Q w^{-\frac{1}{p-1}}\,dx\right)^{p-1} \leq C^p. \tag{3}$$

In particular, $w > 0$ almost everywhere.

Thus, if (2) holds for a weight w, then by Proposition 1, w satisfies (3) for every cube Q, with the same constant C.This gives rise to the next definition by Muckenhoupt [5].

Definition 4 (Muckenhoupt A_p weights)**.** *Let $1 < p < \infty$ and let w be a weight. The A_p* characteristic *of w is*

$$[w]_{A_p} := \sup_Q \left(\frac{1}{|Q|}\int_Q w(x)\,dx\right)\left(\frac{1}{|Q|}\int_Q w(x)^{-\frac{1}{p-1}}\,dx\right)^{p-1}, \tag{4}$$

where the supremum is taken over all cubes $Q \subset \mathbb{R}^d$. We say that $w \in A_p$ if $[w]_{A_p} < \infty$.

Taking the supremum over all cubes in Proposition 1 gives the following.

Corollary 1. *Let $1 < p < \infty$ and let w be a weight that is not almost everywhere zero. If (2) holds with constant C, then $w \in A_p$ and*

$$[w]_{A_p} \leq C^p. \tag{5}$$

Corollary 1 shows that $w \in A_p$ is a necessary condition for (2) to hold. It is natural to ask whether it is also sufficient, that is, whether for every $w \in A_p$ there is a constant C such that (2) holds. Since Corollary 1 moreover shows that any such C must satisfy $C \geq [w]_{A_p}^{1/p}$, we may ask more precisely how large C has to be in terms of $[w]_{A_p}$. The following theorem, due to Buckley [1], answers both questions.

Theorem 1 (Buckley; Theorem C in [4])**.** *Let $1 < p < \infty$. There is a constant $c_{p,d}$, depending only on p and d, such that for every $w \in A_p$ and every $f \in L^p_w(\mathbb{R}^d)$,*

$$\|Mf\|_{L^p_w} \leq c_{p,d}\,[w]_{A_p}^{\frac{1}{p-1}}\,\|f\|_{L^p_w}.$$

The exponent in Theorem 1 cannot be improved.

Proposition 2 (Buckley [1]). *Let $1 < p < \infty$ and $\alpha < \frac{1}{p-1}$. Then there is no constant c such that $\|Mf\|_{L^p_w} \le c\,[w]^{\alpha}_{A_p}\,\|f\|_{L^p_w}$ for all $w \in A_p$ and all $f \in L^p_w(\mathbb{R}^d)$.*

In particular, Theorem 1 shows that $w \in A_p$ is also a sufficient condition, since if $[w]_{A_p} < \infty$, then (2) holds with $C = c_{p,d}[w]^{1/(p-1)}_{A_p}$. Together with Corollary 1, this answers the question posed at the beginning completely and implies the following result.

Theorem 2 (Muckenhoupt). *Let $1 < p < \infty$ and let w be a weight that is not almost everywhere zero. Then M is bounded on $L^p_w(\mathbb{R}^d)$ if and only if $w \in A_p$.*

Theorem 2 is the original result of [5], published in 1972. His proof, as well as Buckley's proof of Theorem 1, relies on the fact that every A_p weight belongs to $A_{p-\varepsilon}$ for some $\varepsilon > 0$, which is a consequence of the reverse Hölder inequality for A_p weights. Buckley's proof moreover uses interpolation. We present neither of these proofs. For the upper bound in Theorem 1 we instead present the elementary proof of Lerner [4], which avoids the $A_{p-\varepsilon}$ property, and as noted above Theorem 2 then needs no separate proof since it follows immidiatly from Corollary 1 and Theorem 1. We complement Lerner's argument with proofs of Propositions 1 and 2.

2 Preliminaries

Lemma 1. *Let Q be a cube with side length $\ell(Q)$, let $y \in Q$, and let Q_y be the cube centred at y with side length $2\ell(Q)$. Then $Q \subseteq Q_y \subseteq 3Q$, where $3Q$ denotes the cube with the same centre as Q and side length $3\ell(Q)$.*

Proof. Every $z \in Q$ satisfies $|z_i - y_i| \le \ell(Q)$ for $i = 1, \dots, d$, since both z and y lie in Q. Hence $Q \subseteq Q_y$. Let c be the centre of Q. Then $|y_i - c_i| \le \frac{1}{2}\ell(Q)$ for $i = 1, \dots, d$, so every $z \in Q_y$ satisfies

$$|z_i - c_i| \le |z_i - y_i| + |y_i - c_i| \le \ell(Q) + \tfrac{1}{2}\ell(Q) = \tfrac{3}{2}\ell(Q)$$

for $i = 1, \dots, d$, that is, $z \in 3Q$. □

Lemma 2. *Let $f \in L^1_{\mathrm{loc}}(\mathbb{R}^d)$. Then $Mf(x) \le 2^d M^c f(x)$ for every $x \in \mathbb{R}^d$.*

Proof. Let $x \in \mathbb{R}^d$. For a cube Q containing x, let Q_x be the cube centred at x with side length $2\ell(Q)$. By Lemma 1, $Q \subseteq Q_x$, and since $|Q_x| = 2^d|Q|$ we obtain

$$Mf(x) = \sup_{Q \ni x} \frac{1}{|Q|} \int_Q |f(y)|\,dy \le \sup_{Q \ni x} \frac{2^d}{|Q_x|} \int_{Q_x} |f(y)|\,dy \le 2^d M^c f(x),$$

where the last inequality holds because each Q_x is a cube centred at x. □

Lemma 3. *Let $1 < p < \infty$ and let $w \in A_p$ be a weight that is not almost everywhere zero. Then $w > 0$ almost everywhere, and $\sigma := w^{-\frac{1}{p-1}}$ is a weight. Moreover $w(Q) > 0$ and $0 < \sigma(Q) < \infty$ for every cube Q.*

Proof. Suppose that $w = 0$ on a set E with $|E| > 0$. Since w is not almost everywhere zero, there is also a set F with $|F| > 0$ on which $w > 0$. Choose bounded subsets $E' \subseteq E$ and $F' \subseteq F$ of positive measure, and a cube Q containing both. Then $w(Q) \ge w(F') > 0$, while $\sigma = \infty$ on E', so $\sigma(Q) = \infty$. Hence the quantity in (4) is infinite for this cube,

which contradicts $w \in A_p$. Thus $w > 0$ almost everywhere, and in particular $w(Q) > 0$ for every cube Q. Since w is locally integrable, $w < \infty$ almost everywhere, so $\sigma > 0$ almost everywhere and $\sigma(Q) > 0$ for every cube Q. Finally, by the definition of $[w]_{A_p}$,

$$\left(\frac{\sigma(Q)}{|Q|}\right)^{p-1} \leq [w]_{A_p} \frac{|Q|}{w(Q)} < \infty$$

for every cube Q. Since every compact set is contained in a cube, σ is locally integrable. □

Lemma 4 ([2, Lemma A.5])**.** *There is a constant N_d, depending only on d, with the following property. Let $E \subset \mathbb{R}^d$ be a bounded set and, for each $x \in E$, let Q_x be a cube centred at x, such that $\sup_{x\in E} \ell(Q_x) < \infty$. Then there is a countable subfamily $\{Q_{x_j}\}_j$ which covers E and satisfies $\sum_j \chi_{Q_{x_j}} \leq N_d$.*

Lemma 5. *Let v be a weight with $v(Q) > 0$ for every cube Q, and let $1 < q < \infty$. Then there is a constant $C_{q,d}$, depending only on q and d, such that*

$$\|M_v^c f\|_{L_v^q} \leq C_{q,d} \|f\|_{L_v^q}$$

for all $f \in L_v^q(\mathbb{R}^d)$.

3 Proof of Theorem 1

In this section we prove Theorem 1, following the argument of Lerner [4].

Proof of Theorem 1. Let $1 < p < \infty$, let $w \in A_p$ and let $f \in L_w^p(\mathbb{R}^d)$. If $w = 0$ almost everywhere, both sides of the inequality vanish, so we may assume that w is not almost everywhere zero. By Lemma 3, $w > 0$ almost everywhere and $\sigma := w^{-\frac{1}{p-1}}$ is a weight with $w(Q) > 0$ and $0 < \sigma(Q) < \infty$ for every cube Q. In particular, the operators M_w^c and M_σ^c are defined.

We first prove a pointwise estimate for Mf. Fix $x \in \mathbb{R}^d$ and a cube Q centred at x, and let $3Q$ be as in Lemma 1. Multiplying and dividing by $w(Q)\,\sigma(3Q)^{p-1}$, we obtain

$$\begin{aligned}
\left(\frac{1}{|Q|}\int_Q |f(y)|\,dy\right)^{p-1} &= \frac{1}{|Q|^{p-1}}\left(\int_Q |f(y)|\,dy\right)^{p-1} \\
&= \frac{w(Q)\,\sigma(3Q)^{p-1}}{|Q|^p} \cdot \frac{|Q|}{w(Q)} \cdot \frac{1}{\sigma(3Q)^{p-1}}\left(\int_Q |f(y)|\,dy\right)^{p-1} \qquad (6)\\
&= \frac{w(Q)\,\sigma(3Q)^{p-1}}{|Q|^p} \cdot \frac{|Q|}{w(Q)}\left(\frac{1}{\sigma(3Q)}\int_Q |f(y)|\,dy\right)^{p-1}.
\end{aligned}$$

Since $w(Q) \leq w(3Q)$ and $|3Q| = 3^d|Q|$, the first factor on the right-hand side satisfies

$$\frac{w(Q)\,\sigma(3Q)^{p-1}}{|Q|^p} \leq 3^{dp}\left(\frac{1}{|3Q|}\int_{3Q} w(y)\,dy\right)\left(\frac{1}{|3Q|}\int_{3Q} \sigma(y)\,dy\right)^{p-1} \leq 3^{dp}\,[w]_{A_p}.$$

For the second factor, let $y \in Q$ and let Q_y be the cube centred at y with side length $2\ell(Q)$. By Lemma 1, $Q \subseteq Q_y \subseteq 3Q$, so

$$\begin{aligned}
\frac{1}{\sigma(3Q)}\int_Q |f(z)|\,dz &= \frac{1}{\sigma(3Q)}\int_Q |f(z)|\,\sigma(z)^{-1}\,\sigma(z)\,dz \\
&\leq \frac{1}{\sigma(Q_y)}\int_{Q_y} |f(z)|\,\sigma(z)^{-1}\,\sigma(z)\,dz \leq M_\sigma^c\big(|f|\sigma^{-1}\big)(y).
\end{aligned}$$

This holds for every $y \in Q$, and therefore

$$\begin{aligned}\frac{|Q|}{w(Q)}\left(\frac{1}{\sigma(3Q)}\int_Q |f(z)|\,dz\right)^{p-1} &= \frac{1}{w(Q)}\int_Q \left(\frac{1}{\sigma(3Q)}\int_Q |f(z)|\,dz\right)^{p-1} dy \\ &\le \frac{1}{w(Q)}\int_Q \Big(M_\sigma^c\big(|f|\sigma^{-1}\big)(y)\Big)^{p-1} w(y)^{-1}\,w(y)\,dy \\ &\le M_w^c\Big(\big(M_\sigma^c(|f|\sigma^{-1})\big)^{p-1} w^{-1}\Big)(x),\end{aligned}$$

where the last inequality holds because Q is centred at x. Inserting both bounds into (6), taking the supremum over all cubes Q centred at x and applying Lemma 2, we arrive at

$$Mf(x) \le 2^d\, 3^{\frac{dp}{p-1}}\,[w]_{A_p}^{\frac{1}{p-1}}\Big(M_w^c\Big(\big(M_\sigma^c(|f|\sigma^{-1})\big)^{p-1} w^{-1}\Big)(x)\Big)^{\frac{1}{p-1}}. \tag{7}$$

We now take norms. Since $1 - \frac{p}{p-1} = -\frac{1}{p-1}$, we have $w^{1-\frac{p}{p-1}} = \sigma$, and similarly $\sigma^{1-p} = w$. The second identity gives

$$\big\||f|\sigma^{-1}\big\|_{L^p_\sigma}^p = \int_{\mathbb{R}^d} |f(x)|^p\,\sigma(x)^{-p}\,\sigma(x)\,dx = \int_{\mathbb{R}^d} |f(x)|^p\,w(x)\,dx = \|f\|_{L^p_w}^p,$$

so $|f|\sigma^{-1} \in L^p_\sigma(\mathbb{R}^d)$. Using the first identity and Lemma 5 with $v = \sigma$ and $q = p$, we obtain

$$\begin{aligned}\int_{\mathbb{R}^d}\Big(\big(M_\sigma^c(|f|\sigma^{-1})(x)\big)^{p-1} w(x)^{-1}\Big)^{\frac{p}{p-1}} w(x)\,dx &= \int_{\mathbb{R}^d}\big(M_\sigma^c(|f|\sigma^{-1})(x)\big)^p\,\sigma(x)\,dx \\ &\le C_{p,d}^p\,\|f\|_{L^p_w}^p,\end{aligned}$$

so the function $\big(M_\sigma^c(|f|\sigma^{-1})\big)^{p-1} w^{-1}$ lies in $L_w^{p/(p-1)}(\mathbb{R}^d)$. Finally, we raise (7) to the power p, integrate against w, and apply Lemma 5 with $v = w$ and $q = \frac{p}{p-1}$. Together with the previous estimate this gives

$$\|Mf\|_{L^p_w}^p \le \Big(2^d\,3^{\frac{dp}{p-1}}\Big)^p\,[w]_{A_p}^{\frac{p}{p-1}}\,C_{\frac{p}{p-1},d}^{\frac{p}{p-1}}\,C_{p,d}^p\,\|f\|_{L^p_w}^p.$$

Taking p-th roots, we conclude that

$$\|Mf\|_{L^p_w} \le c_{p,d}\,[w]_{A_p}^{\frac{1}{p-1}}\,\|f\|_{L^p_w}, \qquad c_{p,d} := 2^d\,3^{\frac{dp}{p-1}}\,C_{\frac{p}{p-1},d}^{\frac{1}{p-1}}\,C_{p,d},$$

and $c_{p,d}$ depends only on p and d. □

References

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